

● EASY

● ALGEBRA

Systems of two linear equations in two variables

03

10 questions · ~15 min

Q1

ALGEBRA

Hiro and Sofia purchased shirts and pants from a store. The price of each shirt purchased was the same and the price of each pair of pants purchased was the same. Hiro purchased 4 shirts and 2 pairs of pants for \$86, and Sofia purchased 3 shirts and 5 pairs of pants for \$166. Which of the following systems of linear equations represents the situation, if x represents the price, in dollars, of each shirt and y represents the price, in dollars, of each pair of pants?

- A. $4x + 2y = 86$
 $3x + 5y = 166$
- B. $4x + 3y = 86$
 $2x + 5y = 166$
- C. $4x + 2y = 166$
 $3x + 5y = 86$
- D. $4x + 3y = 166$
 $2x + 5y = 86$

$$y = 4x - 9$$

$$y = 19$$

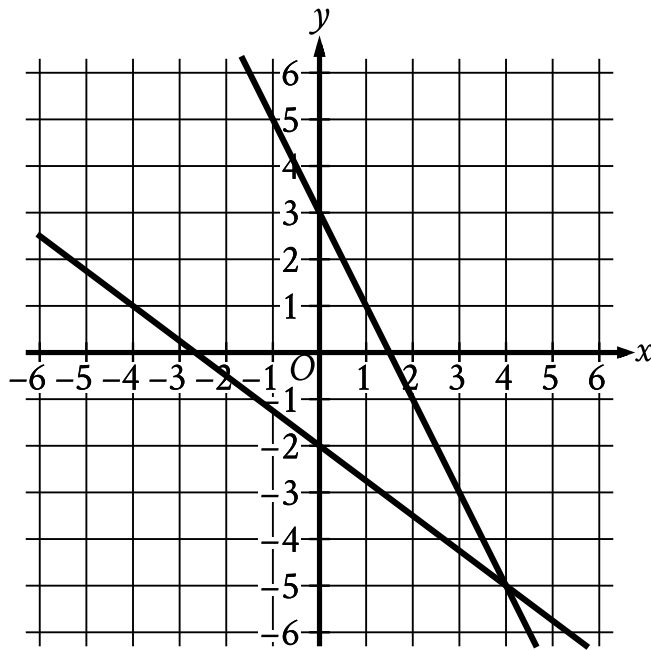
What is the solution x, y to the given system of equations?

A. 4, 19

B. 7, 19

C. 19, 4

D. 19, 7



- For the first line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (0 comma 3)
 - (4 comma negative 5)
- For the second line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (0 comma negative 2)
 - (4 comma negative 5)

The graph of a system of linear equations is shown. What is the solution x, y to the system?

A. 4, -5

B. 0, 3

C. 0, -2

D. -2, 3

$$\begin{aligned}x + y &= 20 \\ 2(x + y) + 3y &= 85\end{aligned}$$

If (x, y) is the solution to the given system of equations, what is the value of y ?

- A. 10
- B. 15
- C. 60
- D. 65

$$y = 12x - 20$$

$$y = 28$$

What is the solution x, y to the given system of equations?

- A. 4, 28
- B. 20, 28
- C. 28, 4
- D. 28, 20

$$10x = 110$$

$$6x - 63 = y$$

The solution to the given system of equations is x, y . What is the value of y ?

A. 63

B. 11

C. 10

D. 3

A petting zoo sells two types of tickets. The standard ticket, for admission only, costs \$5. The premium ticket, which includes admission and food to give to the animals, costs \$12. One Saturday, the petting zoo sold a total of 250 tickets and collected a total of \$2,300 from ticket sales. Which of the following systems of equations can be used to find the number of standard tickets, s , and premium tickets, p , sold on that Saturday?

A.
$$\begin{aligned}s + p &= 250 \\ 5s + 12p &= 2,300\end{aligned}$$

B.
$$\begin{aligned}s + p &= 250 \\ 12s + 5p &= 2,300\end{aligned}$$

C.
$$\begin{aligned}5s + 12p &= 250 \\ s + p &= 2,300\end{aligned}$$

D.
$$\begin{aligned}12s + 5p &= 250 \\ s + p &= 2,300\end{aligned}$$

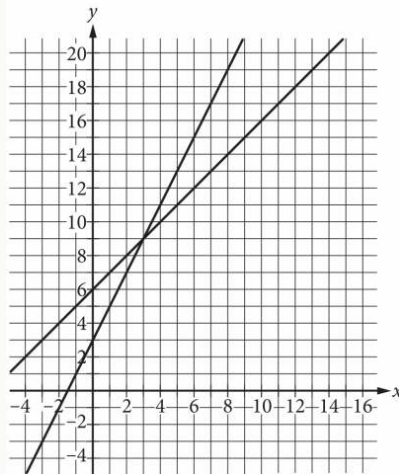
$$x = 4$$

$$y = 5 - x$$

The solution to the given system of equations is x, y . What is the value of y ?

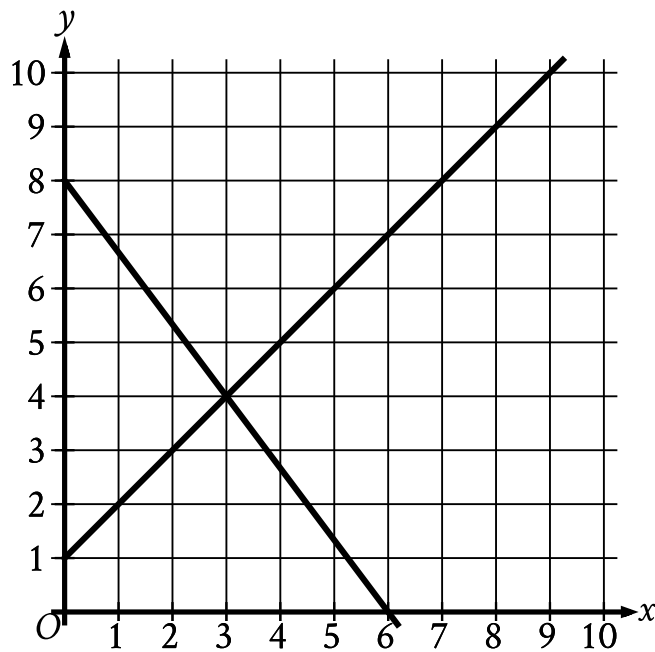
- A. 1
- B. 4
- C. 5
- D. 9

A system of two linear equations is graphed in the xy -plane below.



Which of the following points is the solution to the system of equations?

- A. $(3, 9)$
- B. $(6, 15)$
- C. $(8, 10)$
- D. $(12, 18)$



- For the first line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (0 comma 8)
 - (3 comma 4)
 - (5 comma four thirds)
- For the second line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following points:
 - (0 comma 1)
 - (3 comma 4)
 - (5 comma 6)

The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

A. $(2, 3)$

B. 3,4

C. (4,5)

D. (5,6)